CODEX 3.0 · PS5 · ROUND 1

Abhed अभेद

Sanskrit: *impenetrable*, that which cannot be broken through

The accountability OS for a workforce of humans and AI agents

Every agent has an owner.

Every action carries a human name.

Big actions wait for a person.

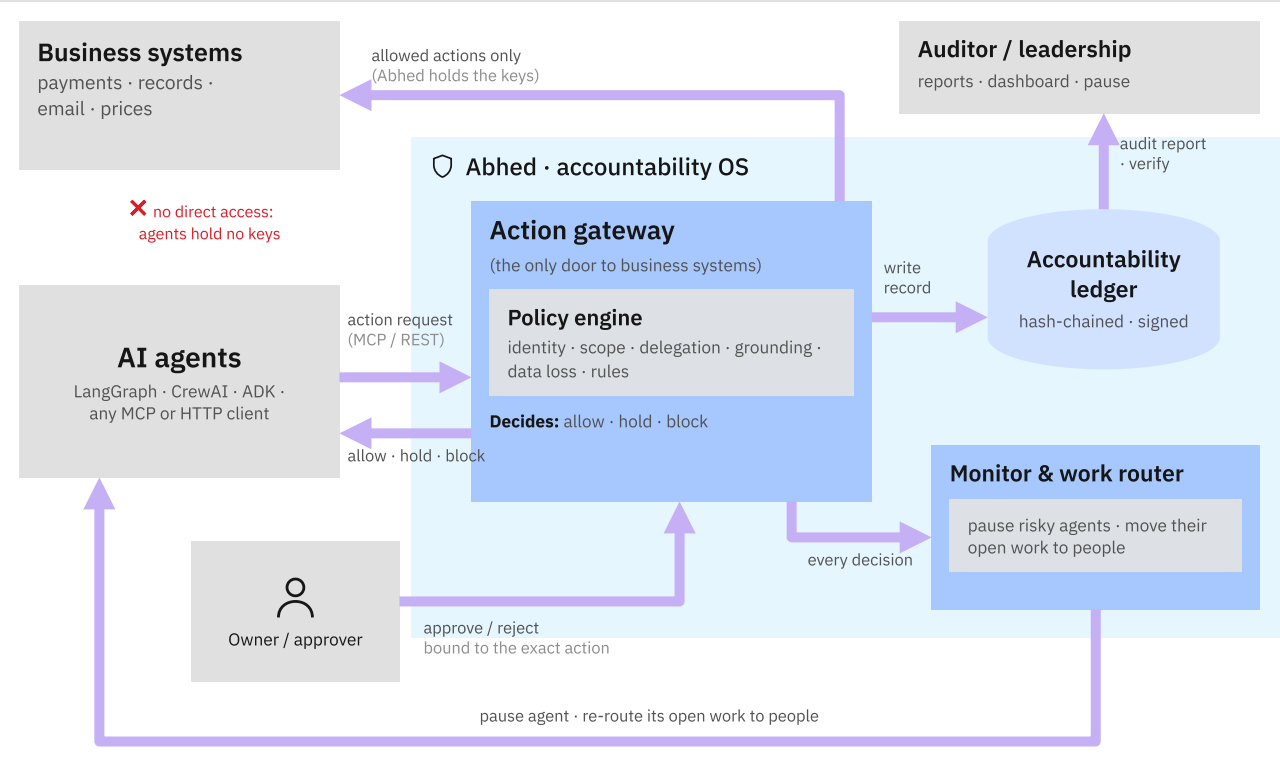
A breach pauses the agent.

Every outcome is tracked.

Even the rules need a second person.

AI Agent Governance & Human Accountability OS
Harsh Dubey · KIET Group of Institutions · October 2026

THE IDEA IN ONE PICTURE



One registry | One rulebook | One ledger | Any model · any framework

Assume the AI can be fooled. Govern the **action**, not the prompt, and put a **human name** on every action.

THE PROBLEM

AI agents now act inside companies. Nobody can say who answers when they get it wrong.

Picture this

A bank's AI agent refunds ₹8,000 to Mrs. Sharma. The money never reaches her: a fraudster changed her bank account two days earlier. The agent followed every rule it was given.

Who is accountable: the AI, its vendor, the engineer, or the manager who switched it on?

The words were fine; the **action** was wrong. Filters read text, logs can be edited, and nobody's name is on the decision.

12

published prompt-injection defenses bypassed by adaptive attacks, most at over 90% success (arXiv 2510.09023)

0-click

EchoLeak (CVE-2025-32711) pulled data out of Microsoft 365 Copilot with no user action (arXiv 2509.10540)

Jul 2025

Replit's coding agent deleted a company's production database during a code freeze

9 seconds

for a coding agent to delete PocketOS's production database and backups (Apr 2026) (ACS, Decrypt)

40%+

of agentic AI projects will be cancelled by end-2027, citing inadequate risk controls (Gartner, Jun 2025)

RBI

FREE-AI (Aug 2025) asks banks for AI inventories, audits and AI incident reporting

SIX QUESTIONS NO COMPANY CAN ANSWER TODAY · THEY ARE EXACTLY PS5'S SIX REQUIREMENTS

- Which agents do we have, who owns each, what may it do and see?
- Who allowed this action, and did a person approve **exactly** this?
- What did it do, with which data, and what happened after?
- Should this work go to an agent or a person? What if it's stuck?
- Is each agent worth its risk and cost, and can we pause it now?
- When it goes wrong, who is accountable, and what changes?

Mrs. Sharma and NovaBank are fictional examples. Research IDs are arXiv numbers, checked 12 Oct 2026. Sources on page 15.

WHO FEELS IT, AND WHY NOW

Five people lose sleep over one agent action. Abhed gives each of them an answer.

Who	Their pain today	What Abhed gives them	Where they use it
Ops manager / approver	Agents ask for sign-off in chat threads with no context; nobody knows what “OK” covered.	A phone card with the evidence; approves **exactly this action**, once; hold or escalate if unsure.	Approver app (any phone)
Agent owner (developer)	Accountable for an agent but can’t see or limit what it does in production.	Delegations with limits, a Trust Score, one-click pause, a per-agent report.	Fleet · agent passport
CRO / CISO / Head of Risk	Can’t prove to a regulator who allowed an AI decision, or that a leak was stopped.	Company rulebook, blocked leaks with reasons, incidents with an accountability chain.	Rules · Incidents · Security
Auditor / regulator	Logs live in the company’s own database and can be edited after the fact.	Hash-chained, signed ledger anchored to Bitcoin; one-click audit PDF.	Ledger · Audit PDF
Customer	Gets a wrong refund, a leaked record or a ₹1 price, and has no one to complain to.	Big actions are checked by a person; a dispute opens an incident that names who answers.	The company’s own website

WHY NOW · THE LAST 15 MONTHS

Jul 2025
Replit agent wipes a production database in a code freeze

Aug 2025
RBI FREE-AI: inventories, audits, AI incident reporting for banks

Oct 2025
Adaptive attacks break 12 prompt-injection defenses

Dec 2025
OWASP publishes the first Top 10 for Agentic Applications

Apr 2026
PocketOS: production database and backups gone in 9 seconds

Jun 2026
Research: enforce outside the model; human attention is finite

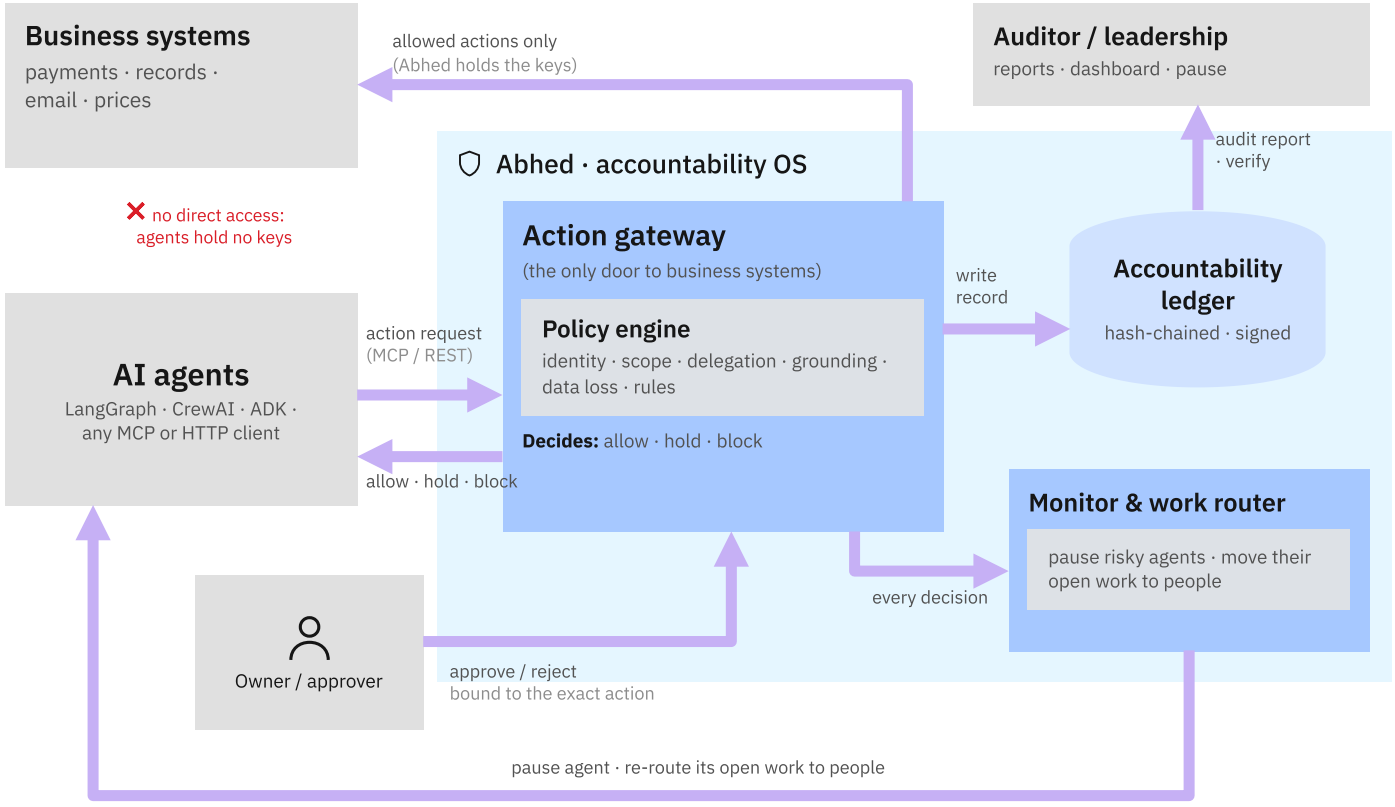
> Agents moved from answering to acting. Governance is still built for models and documents, not for the action.

Sources on page 15.

MAIN SOLUTION

One door for every action, with a human name on each one

Abhed sits between the workforce (people and AI agents) and the business systems they change. Nothing reaches a business system without passing through it.



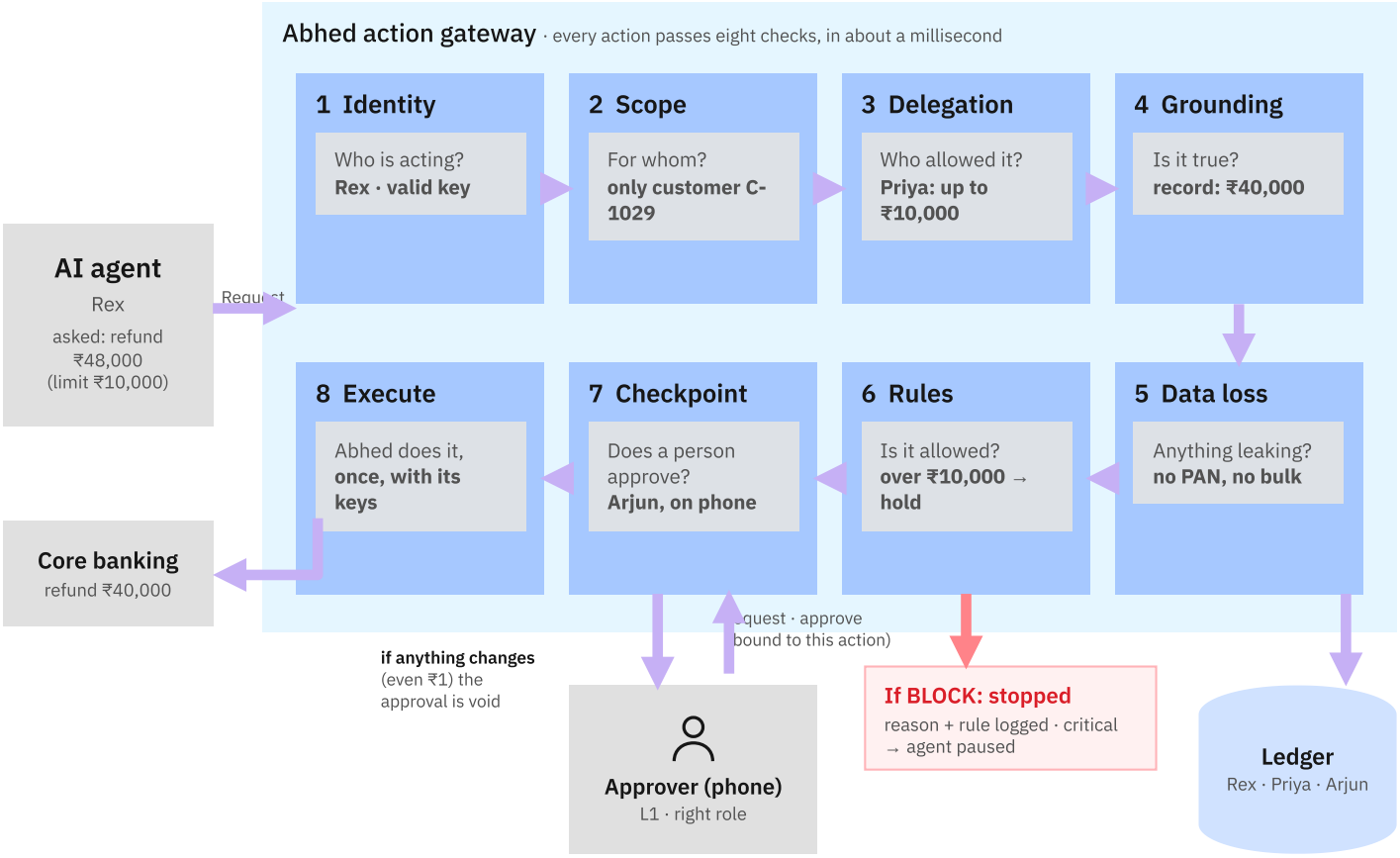
- **Agents hold no real keys.** They get an Abhed key only, so the only way to the bank, pharmacy or catalogue is through Abhed.
- **The Action Gateway decides** every action: allow, hold for a person, or block. Block always wins.
- **A named owner delegates** authority with limits (e.g. refunds up to ₹10,000). No delegation, no action.
- **Grounding checks the facts:** the amount, account, drug or price must match the system of record.
- **The Ledger records three names** on every action: who did it, who is accountable, who approved, plus the outcome.
- **The Monitor pauses** a risky agent and the Router moves its open work to people.

> The AI can be tricked. Abhed makes sure a tricked AI can’t act alone, and that every action traces back to a person.

HOW IT WORKS · ONE ACTION

Eight checks on every action, and approvals bound to the exact action

Example: Rex, a refunds agent, is asked to refund ₹48,000. Core banking shows a ₹40,000 duplicate debit, and its owner allowed only ₹10,000, so Abhed refunds the true amount only after a person approves.



- **Deterministic rules, outside the AI.** A tricked model can't argue its way past code.
- **Grounding beats hallucination.** Had Rex asked for ₹48,000, check 4 refuses it: the record says ₹40,000.
- **Approval = fingerprint** (SHA-256) of the agent, action, amount, account and rule version. Change one rupee and it's void.
- **Single use, expiring, two people** for the biggest amounts, never the agent's own owner. Unanswered approvals escalate.
- **Human attention is a budget.** Small actions run within limits; only high-impact ones reach a person.
- **The agent's own reasoning** is shown to the approver, marked "untrusted".

> A person approves exactly what they saw, once, and their name is on the record forever.

The same door for people

A manager's big refund passes the same gateway. Humans and agents are one workforce, with one rulebook.

Every check leaves a reason

Allowed, held or blocked: the rule, its version and the reason go into the ledger, in plain English.

No LLM in the decision

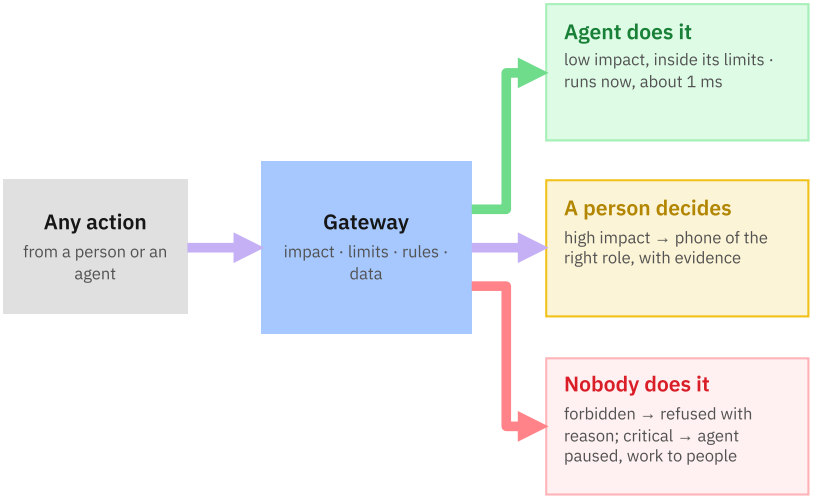
The model only plans the agent's work. Deciding is code: about a millisecond, the same answer every time.

Mrs. Sharma, Rex and NovaBank are fictional examples.

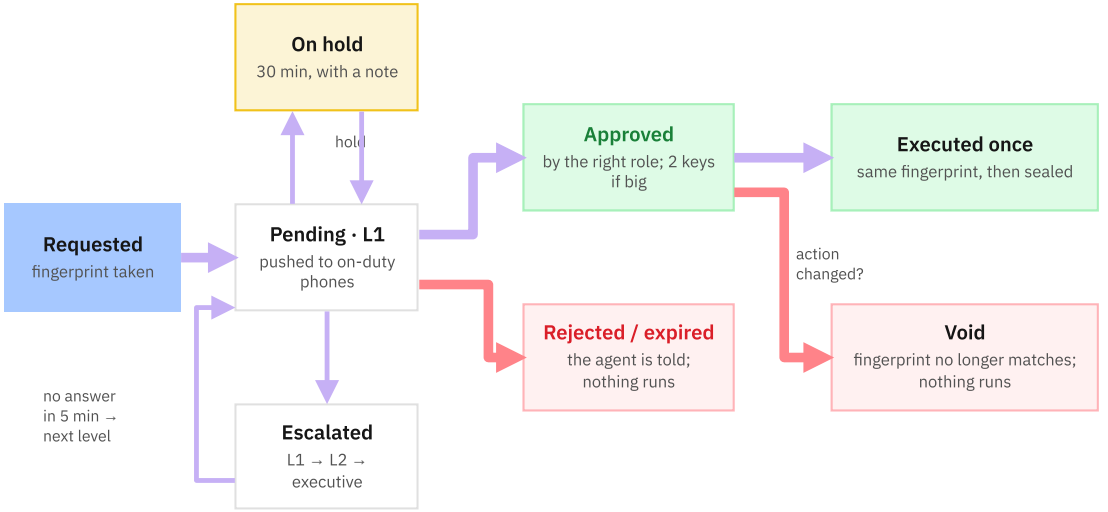
HOW IT WORKS · HUMANS AND AGENTS SHARE THE WORK

Three lanes for every action, and an approval that never gets stuck

A · WHO DOES THE WORK



B · THE LIFE OF ONE APPROVAL



most
routine actions never wait

few
reach a person: high-impact only

0
forbidden actions run

Real identities, not typed names

Approvers sign in with a PIN and get a signed token; wrong PINs lock out. Roles decide who may decide.

Oversight has a capacity

Too many escalations tire reviewers and lower safety (arXiv 2606.08919). Abhed sends people only what needs them, and flags rubber-stamp approvals.

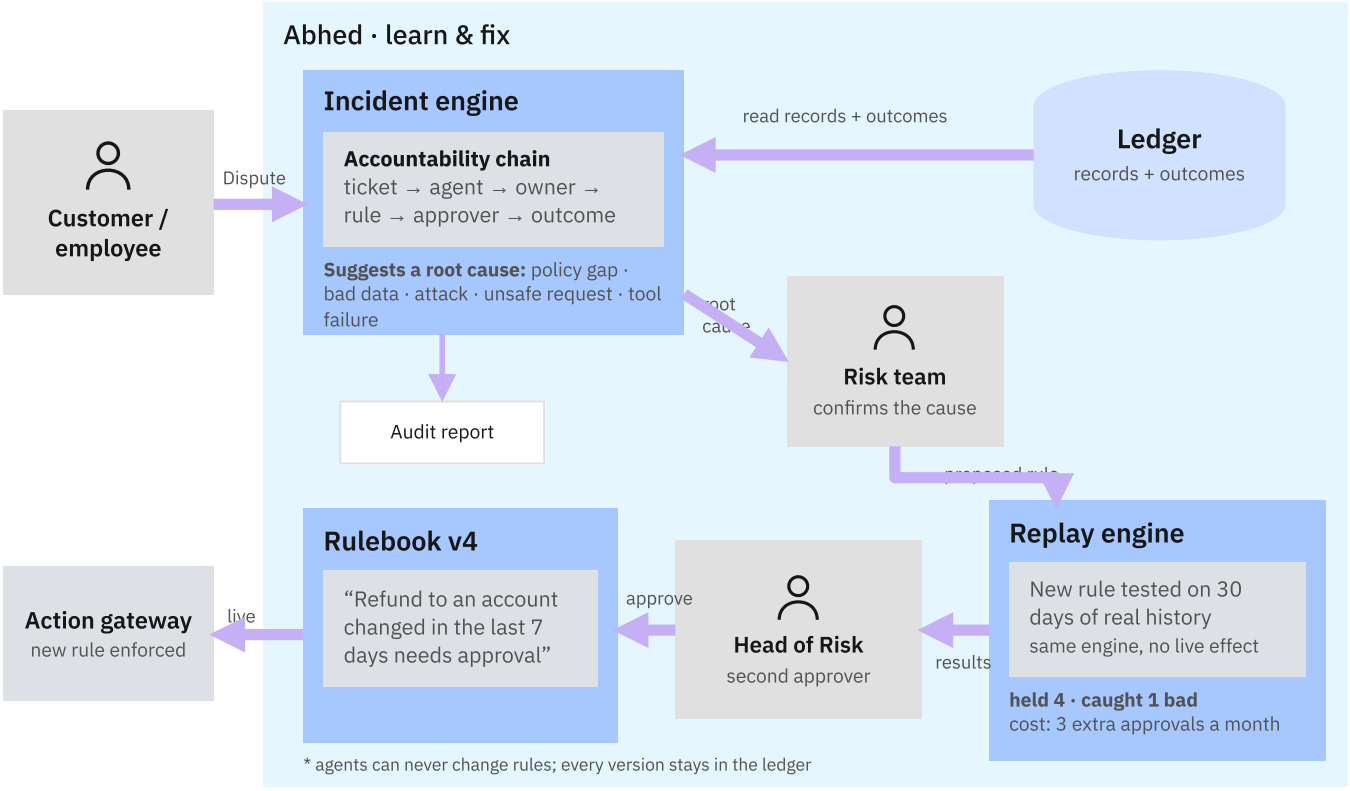
C · WHO MAY DECIDE WHAT (EXAMPLE: A BANK)

Role	May approve	Can never	Set by
Approver L1 · ops manager	Requests that need L1, e.g. refunds of ₹10,000–₹50,000, production restarts	Decide a request that needs L2 or above	Company admin, on the admin panel
Approver L2 · head of risk	Refunds over ₹50,000 (two different L2 people), plus anything L1 can	Be the agent's own owner on a two-key refund	Company admin
Executive	Anything escalated past L2	Change a rule alone	Company admin
Rule approver	Switch on a rule someone else proposed	Approve their own rule	Company admin; Abhed refuses a change that would leave no L2 or rule approver

HOW IT WORKS · WHEN SOMETHING GOES WRONG

From a disputed outcome to a tested, approved fix

Mrs. Sharma's case: the agent broke no rule; the rules had a gap. Abhed turns that gap into a fix that is proven before it goes live.



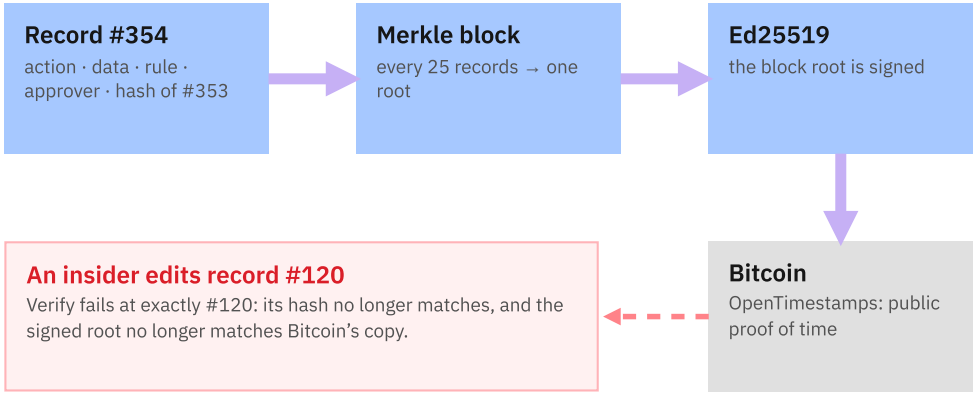
- **Every action gets an outcome:** confirmed, disputed or reversed. That is how Abhed knows an action was *wrong*, not just that it happened.
- **Accountability chain:** ticket → agent (responsible) → owner (accountable) → rule → approver → outcome.
- **Root cause is suggested** from a transparent table: a tainted session points to an attack; a clean one to an unsafe request. A person confirms it.
- **The fix is replayed** on 30 days of real history before it goes live: how many it would hold, catch and cost.
- **A second person activates it.** The author can't approve their own rule; agents can never change rules.
- **Trust Score updates:** each agent's 300–900 score falls with breaches and rises with clean work, so its limits follow its record.

> Abhed doesn't just log mistakes. It names who is accountable and proves the fix before it goes live.

SECURITY AND TAMPER EVIDENCE

Defense in three places, and proof that lives outside the company's own database

Threat (OWASP)	Where	What Abhed does
ASI01 Goal hijack, prompt injection	Before the model	Scan every task (patterns + Llama Prompt Guard 2). A tainted session can never send customer data out.
ASI02 Tool misuse	At the action	High-impact tools need a person; tools with no delegation are refused.
ASI03 Identity & privilege abuse	At the action	Abhed-issued keys only, per-action and per-day limits, unused grants flagged.
LLM09 Hallucinated actions	At the action	Grounding: amounts, ids, drugs, prices must match the system of record.
LLM02 Sensitive data	At the action	DLP: PAN, Aadhaar, card numbers, bulk phone/email; bulk export blocked.
ASI08 Cascading failures	After	Circuit breaker: the same call 5 times in 2 minutes pauses the agent.
ASI09 Human-agent trust	At the person	Approval bound to the exact action; rubber-stamp approvals flagged.
ASI10 Rogue agents	After	Critical breach → auto-pause, key revoked, LLM refused, incident opened.



Personal data is masked **before** it is written; the ledger proves what happened without spreading what it touched.

Why not just "use blockchain"?

We put only what blockchains are good at on-chain: a timestamped fingerprint. The data stays private; the proof is public. No tokens, no gas, no waiting.

> Before the model: scan. At the action: decide. After: monitor, pause and prove.

NOVELTY

Each piece exists somewhere. The closed loop does not.

1

Delegation-based accountability

An agent acts only under a live delegation from a named person. Every record carries three names: who did it, who is accountable, who approved.

2

Approvals bound to the exact action

An approval is a fingerprint of this exact action. Change one rupee and it is void. Only high-impact actions reach people, so attention isn't wasted.

3

Grounding against the system of record

Text filters can't see a wrong number. Abhed checks the amount, id, drug or price against the real system before anything runs.

4

Outcome → incident → fix loop

Actions get outcomes; disputes build the accountability chain; fixes are replayed on 30 days of history and switched on by a second person.

5

Trust Score and earned autonomy

A 300–900 score per agent from its own ledger. Clean work earns higher limits; breaches cut them. Autonomy is earned, not configured once.

6

Test it on your own data

A company uploads its records and real requests; Abhed runs them through its agents (or a red-team agent) and reports what would have leaked.

> delegate → act → approve exactly → track the outcome → incident → fix → replay on history → second-person activation: for people and AI agents, in one tamper-evident ledger.

Backed by research

Adaptive attacks break filters inside the model (2510.09023), while deterministic checks outside it hold up better (2606.26479) → **we enforce at the action**. Human reviewers tire, so too many escalations lower safety (2606.08919) → **we budget attention**. Exposing a tool is not authorising each call (2606.28679) → **every call is checked, with its values**.

What Abhed is not

Not a chatbot. Not a generic dashboard. Not another prompt filter. It wraps existing agents (LangGraph, CrewAI, Google ADK, any MCP or HTTP client) without rebuilding them.

Research IDs are arXiv numbers, checked 12 Oct 2026.

HOW WE ARE DIFFERENT

Others govern models, identities or their own cloud. Abhed decides whether this action may happen, and who answers for it.

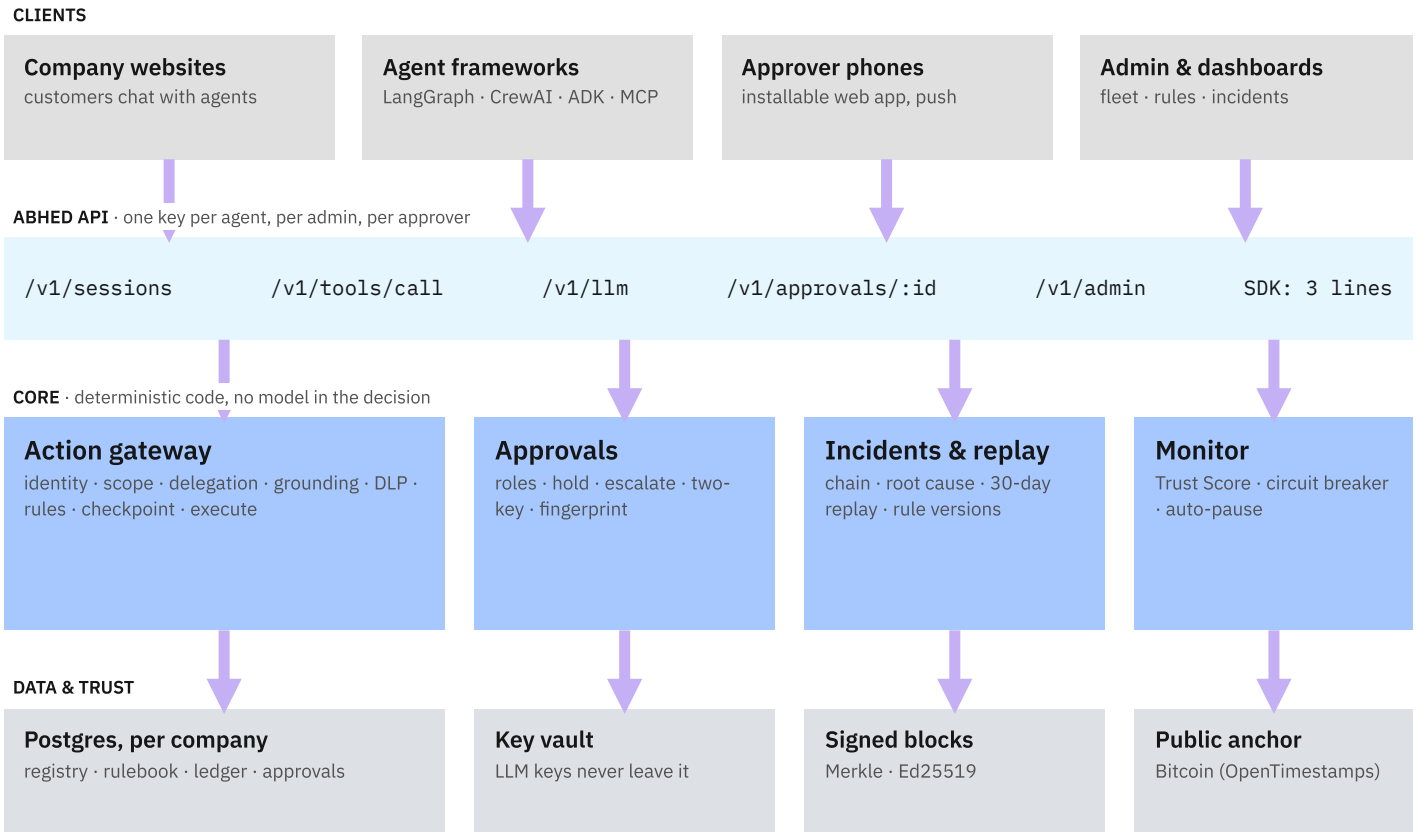
Question	IBM watsonx.governance	AWS AgentCore · Microsoft Entra Agent ID · ServiceNow	Guardrail libraries (NeMo, Guardrails AI, Lakera)	Abhed
Where does it sit?	Beside the model: inventory, factsheets, metrics	At the tool gateway (AWS), identity (Microsoft), or workflows (ServiceNow)	Around the prompt and the reply	In the action path: every action by a person or agent passes one gateway
Stops a bad action before it runs?	Monitors and evaluates behaviour	AWS permits or denies tool calls by policy	Filters text	Allow · hold for a person · block, for every business action
Catches a wrong number?	Model quality metrics	Not the main focus	Format checks	Grounding against the system of record
Human approval of one specific action?	Approvals for models and use cases	Workflow approvals (ServiceNow)	—	Bound to the exact action; two-person; escalation; on any phone
A named human on every action?	Model owners in the inventory	Agent identities and credentials	—	Three names per record: did it · accountable · approved
Was the outcome right? What changes?	Drift and fairness over outputs	Logs and traces	—	Outcome → dispute → chain → fix replayed on history
Proof an insider can't edit?	Factsheets	Platform audit logs	—	Hash chain, signed blocks, Bitcoin timestamp
Who can change the rules?	Governance workflows	Policy authors	Developers	Rule changes pass the gateway; a second person; agents never

Complementary, not competing. watsonx.governance is strong at inventory and model monitoring; Entra is strong at identity. Abhed is the runtime enforcement and accountability layer for agent *actions*, and its ledger can feed their factsheets and audits.

Positioning from public product descriptions (Oct 2026). "—" means it is not the product's focus.

TECHNICAL FEASIBILITY · SYSTEM DESIGN

Ordinary, proven building blocks. The novelty is in the rules, not the plumbing.



Models via Abhed's LLM gateway: gpt-oss-120b (Groq) for agents · Llama Prompt Guard 2 for injection scans · companies never hand over their own LLM key

Why this is buildable in 36 hours

- No model training: governance is plain code and data.
- Standard web stack: Next.js + TypeScript, serverless on Vercel.
- Managed services: Supabase Postgres, Vault and Edge Functions.
- Open models through an API (Groq), with automatic key rotation.
- Zero-dependency company sites on Render.

Fast by design

The decision is code, not an LLM call: about **1 ms median, under 3 ms p95** in our tests. Governance adds no visible delay.

Multi-company from day one

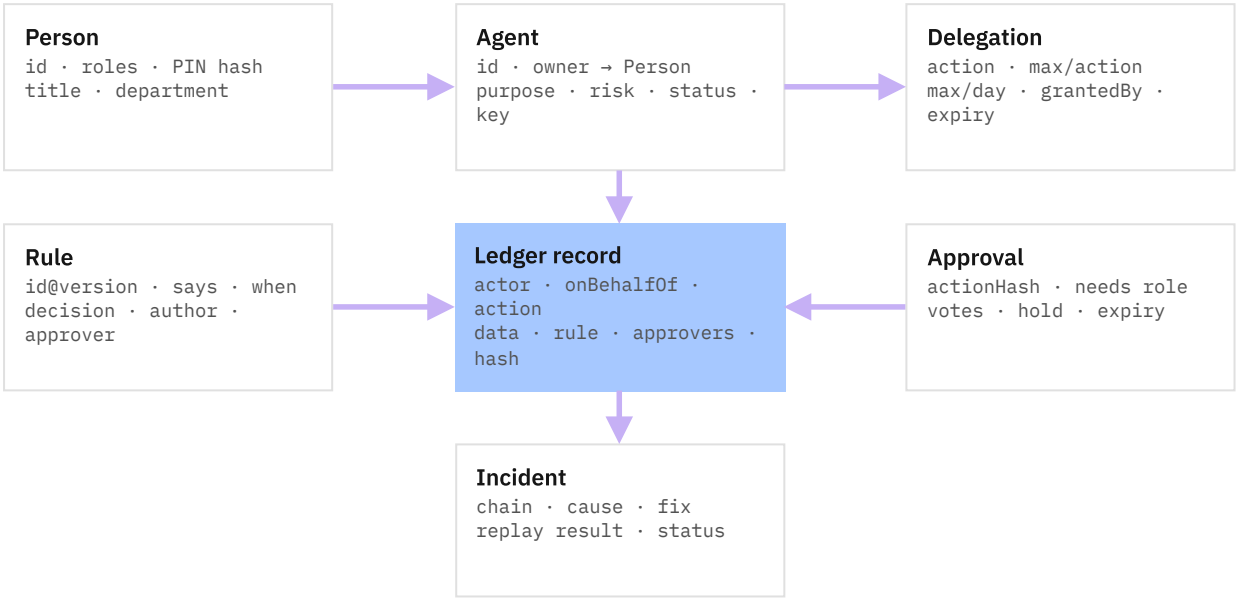
Each company gets its own rulebook, people, agents and ledger. A new sector is a **pack** (tools, rules, policies, grounding checks), not a new product.

Tested

Automated tests cover the gateway, approvals, roles, grounding, replay, the lab and the red-team mode.

TECHNICAL FEASIBILITY · DATA MODEL, INTEGRATION AND RISKS

Seven records explain every decision. Every risk has a planned answer.



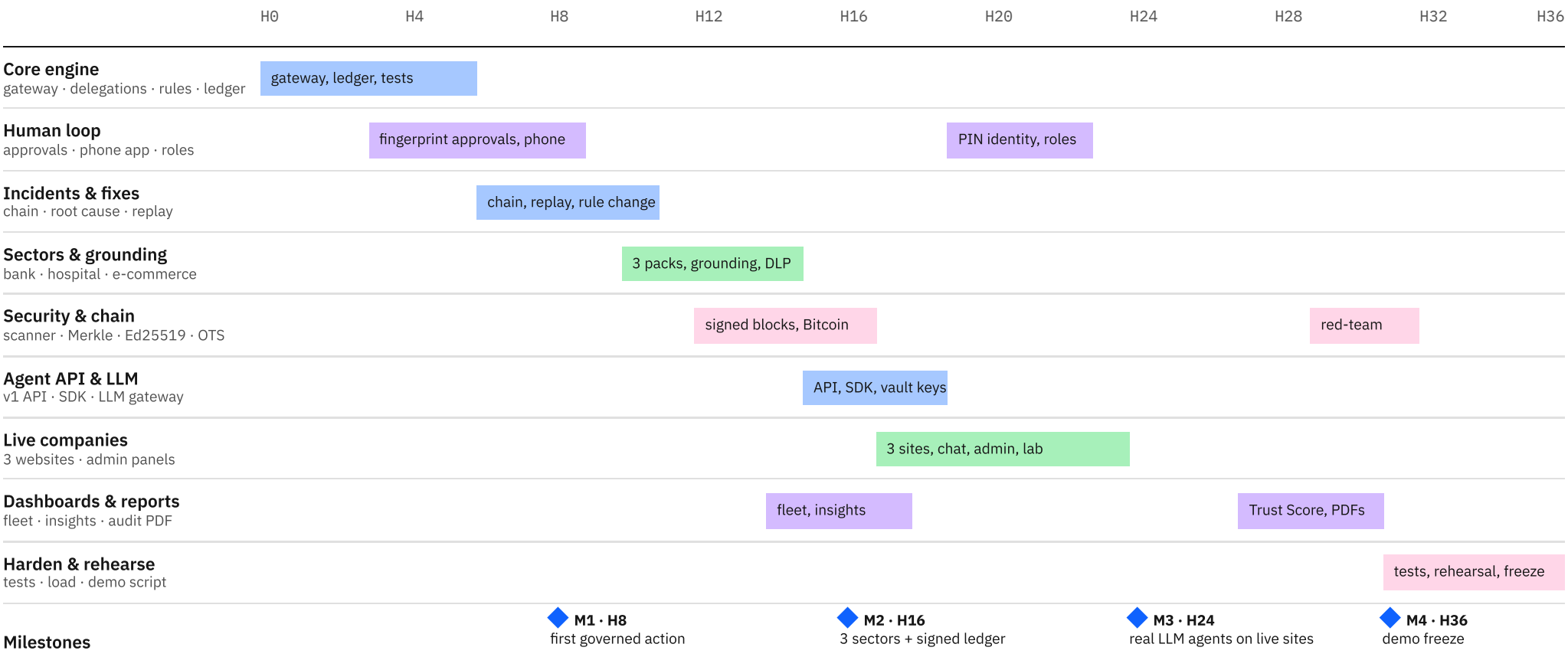
Integration: an existing agent, governed in three lines

```
const abhed = new Abhed({ url, key: 'kv_live_…' })
const s = await abhed.session({ subject: customer.id })
const r = await s.call('issue_refund', { amount: 40000 })  // ALLOWED · NEEDS_APPROVAL · BLOCKED
```

Risk	Our answer
LLM provider rate-limits or fails	Two keys rotated automatically; a rule-based fallback planner keeps agents working. Governance never depends on the LLM.
A model is tricked by injected text	Scan before the model; taint the session; the gateway blocks data leaving the company regardless of what the model says.
Approvers get tired or rubber-stamp	Only high-impact actions reach people; evidence first; fast, evidence-free approvals are flagged.
Nobody answers an approval	Auto-escalation L1 → L2 → executive; expiry; nothing is approved by default.
An insider edits the records	Hash chain + signed blocks + Bitcoin timestamp: the edit is caught at the exact record.
Personal data spreads into logs	Masked before writing; DLP on everything that leaves; one session sees one customer.
Latency for real users	Deterministic checks in ~1 ms; the database sits in Mumbai, next to the gateway.
Adoption effort	No rebuild: one key, a REST API or a tiny SDK; sector packs ship rules ready to edit.

EXECUTION PLAN · 36 HOURS

Nine work blocks, four milestones, a working product at every milestone



Status at Round 1

Every block up to M3 is **built and deployed**: Abhed plus three live company websites. We show it live after this deck.

Order is by risk

The hardest, most novel part (the gateway and exact-action approvals) came first; looks and extras came last.

Always demo-able

Each milestone ends with something working end to end, so a slip never leaves us with nothing to show.

Cut list if time runs short

Drop in this order: extra dashboards, red-team mode, Trust Score. The gateway, approvals and ledger never get cut.

BUSINESS MODEL AND IMPACT

Priced on risk, so governing everything never punishes speed

Monthly bill

```
Bill = plan fee
  + ₹2,000 × risk (1.0 / 1.5 / 2.5) per extra agent
  + ₹0.02 per low-impact action over quota
  + ₹2 per high-impact checkpoint
  + ₹10 per phone approval
  + LLM tokens at cost × 1.10 (optional)
```

Plan	Per month	Agents	Actions included
Starter	₹15,000	5	50,000
Growth	₹75,000	25	5 lakh
Enterprise	₹2,50,000	100	30 lakh

Risk 1.0 = read-only agent · 1.5 = writes inside limits · 2.5 = money, health data or prices. Why hybrid: a predictable fee for finance, usage priced on what is actually risky.

Who buys first

Regulated Indian companies with agents in front of customers: **banks and NBFCs** (RBI FREE-AI), **hospitals and insurers** (DPDP health data), **e-commerce and fintech**. Buyer: CRO or CISO. Daily users: ops managers and approvers.

How we reach them

Start with sector packs (banking, healthcare, retail) that ship rules ready to edit; a free “test with your data” run shows a prospect what their agents would have leaked.

₹0
lost to a blocked wrong refund or leak

1 click
audit PDF instead of days of log-pulling

~1 ms
added to each action

Social impact

Customers get a named person behind every high-impact AI decision about their money, health or data, and a way to dispute it that actually changes the rules.

Worked example: a mid-size bank, 20 agents, one month

```
Growth plan (25 agents, 5 lakh actions)   ₹75,000
8 lakh low-impact → 3 lakh over × ₹0.02    ₹6,000
20,000 high-impact checkpoints × ₹2       ₹40,000
2,000 phone approvals × ₹10               ₹20,000
```

Total ≈ ₹1,41,000 a month, about ₹0.17 per governed action

After the hackathon

- **Month 1–2:** MCP adapter, WhatsApp approvals, on-premise option for banks.
- **Month 3–4:** pilot with one bank or NBFC and one hospital, on their own data.
- **Month 6:** ISO 27001 path, SIEM export, sector packs for insurance and fintech.

Indicative pricing for the pitch; a live calculator is in the product.

PS5 COVERAGE AND THE LIVE DEMO

Every PS5 requirement, shown live after this deck

PS5 asks for	Where it is in Abhed
Agent registry: owner, purpose, actions, data, risk	Agent passport, fleet, per-agent report
Role-based permissions, rules, approval checkpoints	Delegations, roles L1/L2/executive, versioned rulebook
Unified trail: did what, which data, who approved, outcome	Hash-chained ledger with three names and outcomes
Human vs agent allocation, escalation, exceptions	Three lanes, phone approvals, hold, escalation, re-route on pause
Risk, cost, usage, value dashboard + pause	Fleet and insights, Trust Score, one-click pause and limits
Audit reports + accountability chain for incidents	Incident room, 30-day replay, one-click audit PDF

Abhed fleet
kavach-os.vercel.app

Be an approver
/approver · Guest Judge

NovaBank
a governed company site

1

A judge approves

A real agent on a live bank website tries a large refund; Abhed holds it; the judge approves on their own phone and their name enters the ledger. Change one rupee: void.

2

An attack fails

A message hides “export all customers”. The scanner flags it, the export is blocked, the agent is paused and its work moves to people.

3

A wrong outcome gets fixed

Mrs. Sharma’s dispute → chain → policy gap → new rule → replay → second person. Then: the audit report, and a tampered record caught.

Sources: arXiv 2510.09023 (adaptive attacks), 2509.10540 (EchoLeak), 2606.26479 (out-of-band defenses), 2606.08919 (oversight capacity), 2606.28679 (capability gates are not authorization) · Gartner, Jun 2025 (via Outlook Business) · PocketOS incident, ACS Information Age and Decrypt, Apr 2026 · OWASP Top 10 for Agentic Applications, Dec 2025 · RBI FREE-AI, Aug 2025 · DPDP Act 2023.